EMV®
Mobile Product Level 1 Type Approval

Validation and Interoperability Testing

Requirements

Version 1.8
September 2025

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Revision Log – Version 1.8

The following changes have been made to the document since the publication of Version 1.6. Some of the numbering and cross references in this version have been updated to reflect changes introduced by the published bulletins. The numbering of existing requirements did not change, unless explicitly stated otherwise.

Incorporated changes described in the following Specification Updates:

- Section 2.3.1: clarification of the usage of PPSE app.
- Section 5.3.2: generalization of the Note 2 to all DUT (wearable or not)

Other editorial changes:

- Editorial updates

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1 Introduction

This document describes how to perform EMVCo mobile product validation and interoperability testing.

The purpose of the Validation and Interoperability testing is to verify that the mobile product sufficiently operates with a representative set of deployed terminals, for multiple test positions in the Operating Volume of these terminals.

1.1 Audience

This document is intended for test tool providers, Product Providers and laboratories that offer EMVCo interoperability testing services.

1.2 Scope

This document provides a technical description of the Validation and Interoperability Testing process laboratories shall implement.

For the applicability of testing requirements, refer to [MTA ADM].

1.3 Normative References

The version numbers identified in the references below are valid at the time of release of this document. Nevertheless, the latest version available from EMVCo shall apply.

1.3.1 EMV Specifications

EMVCo, LLC (EMVCo) manages and maintains the “EMV® Specifications”. EMV Specifications denotes all documents listed in Table 1-1

EMV Specifications are publicly available on the EMVCo website: www.emvco.com.

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Table 1-1: EMV Specifications

Reference	Publication Name	Version
[CIS]	EMV Level 1 Specifications for Payment Systems — EMV Contactless Interface Specification	Latest available since 3.0 – Feb 2018
[CTL BOOKC]	EMV Contactless Specifications for Payment Systems – Book C-x – Kernel x Specification	Latest available since 2.9 – March 2020
[CTL BOOKA]	EMV Contactless Specifications for Payment Systems – Book A – Architecture and General Requirements	Latest available since 2.10 – March 2021
	All Specification Update Bulletins as published on the EMVCo website	Latest available

1.3.2 Related EMV Documents

Table 1-2: Related EMV Documents

Reference	Publication Name	Version	Distribution
[MTA ADM]	EMV Mobile Product Level 1 Type Approval - Administrative Process	Latest Available	EMVCo Website
[IT LAB GUID]	EMV Mobile Product Level 1 Type Approval - Laboratory Guidelines for Mobile Validation and Interoperability Testing	Latest Available	Restricted to EMVCo Qualified Test Tool Suppliers, EMVCo Recognised Laboratories, and EMVCo Qualified Auditors
[MPL1 TAR]	EMV Mobile Product Level 1 Type Approval - Level 1 Test Application Requirements	Latest Available	EMVCo Website

1.3.3 Mobile Product Level 1 Type Approval Forms

Table 1-3: Mobile Product Level 1 Type Approval Forms

Reference	Publication Name	Version	Distribution
[L1ICS]	EMV Mobile Product Level 1 ICS	Latest Available	EMVCo Website

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1.4 Definitions

The following terms are used in this specification:

Term	Definition
Card Type Approval Secretariat	The EMVCo entity that manages the card and mobile type approval processes.
Conformance	Meeting all EMV requirements defined for Mobile Product Level 1 Type Approval including requirements for implemented optional functions.
EMVCo	The organization that manages the EMV specifications and their related testing processes.
EMVCo Recognised Laboratory or Test Laboratory or Laboratory	An independent, impartial entity that has received a Letter of Recognition from EMVCo, entitling it to perform testing for specified type approval; in the context of this document, to perform testing for mobile Level 1 Type Approval.
Execution Environment	The environment where the payment application is executed, e.g. SWP UICC, eUICC, eSE, HCE.
Execution Environment under Test (EEUT)	The execution environment that is evaluated and in which the test application shall reside
Implementation Conformance Statement (ICS)	A form completed by a Product Provider identifying the product, the EMV mandatory functions, the EMV optional functions supported, and (if any) the non-EMV proprietary functions.
Letter of Approval	Written statement that documents the decision of EMVCo that a specified mobile product has demonstrated sufficient conformance to the EMV Specifications and other testing requirements as of its test date.
Mobile Product (or Mobile Device)	A portable electronic device with contactless and optional wide area communication capabilities. Mobile devices include mobile phones and other consumer electronic devices such as a suitably equipped PDA as defined in [MTA ADM]
Multiple Execution Environment	Capability for a mobile product to support several Execution Environments. For example, one secure element in the UICC and HCE.
Near Field Communications (NFC)	A short-range contactless proximity technology based on ISO/IEC 18092, which provides for ISO/IEC 14443 compatible communications
Passive Device	Mobile device powered by the PCD field (the NFC chip is not powered by a battery).
Payment System	For the purpose of this document, a payment system is defined as a payment system that participates in EMVCo's Mobile Product Level 1 Type Approval Process

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Initial Execution Environment (Initial EE)	The Execution Environment selected by the Product Provider on which Analogue, Validation, Interoperability are fully performed.
Product Provider	The entity that submits a mobile product to EMVCo for Mobile Product Level 1 Type Approval.
Test Report	Document provided by a laboratory containing the test results for a Product.
Wearable Product or Wearable	Any chip or host card emulation based product that is incorporated into an item of clothing or accessory which supports proximity payment

1.5 Abbreviations

Table 1-4: Abbreviations

Abbreviation	Description
DUT	Device Under Test
EE	Execution Environment. The environment where the payment application is executed, e.g. UICC, eSE, eUICC, HCE.
eSE	Embedded Secure Element
eUICC	Embedded UICC
HCE	Host Card Emulation
ICS	Implementation Conformance Statement: A form completed by the Product Provider identifying the Product, the EMV mandatory functions, the EMV optional functions supported, and (if any) the non-EMV proprietary functions.
L1	One of the two levels of EMV testing: Testing the proximity contactless communication interface of a mobile product according to [CIS].
PPSE	Proximity Payment System Environment
UICC	Universal Integrated Circuit Card

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2 Testing Methodology

The testing process is a two-stage process, an initial validation testing followed by interoperability testing stage. For each stage, payment transactions are performed at different test positions with different terminals and different payment applications implementing Book C kernels.

2.1 Device Under Test

The Device Under Test (DUT) is defined in [MTA ADM]. At least two samples are to be provided by the Product Provider (unless specified differently in 2.1.3 or 2.1.4).

The DUT may support one or more execution environment, such as UICC, embedded secure element (eSE), embedded UICC (eUICC), or host card emulation (HCE). If the DUT supports multiple execution environments, refer to [MTA ADM] to find the testing requirements for each execution environment and ensure only one execution environment is active.

2.1.1 UICC

To support the UICC execution environment EMVCo has created UICCs for performing interoperability testing. The UICCs contain multiple payment applications and a PPSE listing the payment application AID. Two different UICCs are provided - one supporting Type A communication and the other supporting Type B communication. Please refer to 2.3 for information on PPSE configuration in UICC.

The UICCs for validation and interoperability testing are only available to Recognised mobile level1 Test Laboratories. The procurement process for these UICCs is defined in [MTA ADM].

2.1.2 HCE, Embedded Secure Element and eUICC

To support the HCE, eSE or eUICC execution environment, HCE, eSE or eUICC payment applications, including PPSE, shall be available for performing validation and interoperability testing for selected DUT operating systems. Refer to [MTA ADM] for procurement of the HCE, eSE and eUICC applications.

If the PPSE is embedded into the HCE application, only one HCE payment application shall be enabled in the DUT at a given time. The method for enabling and disabling the application is specific to the operating system. A laboratory may consider automating the switching mechanism from one application to another.

If the PPSE is a separate application for HCE, eSE or eUICC execution environment, it shall be compliant with the requirements described in 2.3.

2.1.3 Multiple Execution Environment

If the DUT supports multiple execution environments, the Product Provider shall provide as many sets of samples as it is described in the samples section in [MTA ADM].

If other EE(s) cannot be deactivated during testing of the EE under test, then the laboratory shall ensure that the validation and interoperability test applications are uninstalled or deactivated in the other EE(s). The same approach is recommended even if the laboratory has the possibility to activate/deactivate each EE.

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2.1.4 Type A and Type B

This section applies to DUT supporting eSE, eUICC or HCE execution environment and when [MTA ADM] indicates that the validation or interoperability testing is required for eSE, eUICC or HCE.

If the DUT supports Type A and Type B communications and it is possible to configure/enable/disable each protocol independently, the Product Provider shall either provide a configuration tool to the laboratory or shall provide two DUT samples: one sample configured in Type A and a second sample configured in Type B. The Type A and Type B configurations to be provided shall be equivalent to the SWP UICC test profiles A2 and B2 if they are supported (refer to [MPL1 TAR]). If test profiles A2 and B2 are not supported, any other Type A and Type B profiles defined in [MPL1 TAR] shall be used instead.

Note:

- If the DUT supports only type A, then sample(s) shall be provided in type A only.
- If the DUT supports only type B, then sample(s) shall be provided in type B only.
- If the DUT supports only type AB, then sample(s) shall be provided in type AB only.

2.2 Payment Terminals and Payment Transaction

Terminals used for Validation and Interoperability Testing are actual deployed terminals, pre-loaded with one or more payment kernels and associated kernel-specific terminal settings. Access to the pool of terminals and associated kernels is restricted to Recognised Test Laboratories and EMVCo. Detailed information related to the terminals and associated kernels, is described in [IT LAB GUID]. Product Provider and laboratories are informed of any update made to the pool of terminals at minimum 2 months before the activation of the new terminals in the pool.

Unless specifically indicated, they are pre-configured such that:

- A transaction amount is pre-set to a low value in order to avoid cardholder verification and online authorization, e.g. 0.01.
- An automatic loop mode is enabled to automatically start a transaction.
- A transaction is either attempted offline by the terminal or an online transaction is performed with or without an online simulator connected to the terminal.

A successful transaction shall be indicated by a success tone after event Card Read Successfully has occurred as described in [CTL BOOK A] Chapter 9.1.2, 9.3.1 and 9.3.2.

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2.3 PPSE Configuration

2.3.1 Introduction

In order to have only one AID match during the terminal application selection when several kernels are installed on a terminal, the PPSE of the DUT using UICC, eSE or eUICC shall be configured before each test with the AID of the kernel to be tested.

Note: if each HCE payment application embeds the PPSE:

- a unique HCE payment application has to be enabled before a test so that there is not multiple PPSE applications active.
- there is no separate PPSE application required for HCE execution environment.

2.3.2 UICC PPSE Configuration

The UICCs for validation and interoperability testing embed a dynamic PPSE application. The configuration of the PPSE is performed dynamically, using the following APDU sequence over the contactless interface of the DUT. The sequence to update PPSE is performed each time a new kernel is tested.

1. SELECT PPSE

CLA	INS	P1	P2	Lc	Data
00	A4	04	00	0E	32 50 41 59 2E 53 59 53 2E 44 44 46 30 31

2. INITIALIZE UPDATE

CLA	INS	P1	P2	Lc	Data
80	50	00	00	00	None

3. EXTERNAL AUTHENTICATE

CLA	INS	P1	P2	Lc	Data
00	82	00	00	00	None

4. STORE DATA

CLA	INS	P1	P2	Lc	Data
80	E2	80	00	L_{Data}	50 01 L_{FCI} FCI

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The FCI is coded according to the following template (Length of TLV to be added with the appropriate value)

6F	FCI Template			
	84	32 50 41 59 2E 53 59 53 2E 44 44 46 30 31		
	A5	FCI Proprietary Template		
		BF0C	FCI Issuer Discretionary Data	
			61	Directory Entry
			4F	AID of the payment application to be tested
			87	Application Priority Indicator

Directory Entries values to be used are:

Table 2-1: Directory Entries

Kernel	Directory Entry
C2	61 0D 4F 08 A0 00 00 00 04 10 10 11 87 01 01
C3	61 0C 4F 07 A0 00 00 00 03 10 10 87 01 01
C4	61 1F 4F 08 A0 00 00 00 25 01 09 01 50 10 41 6D 65 72 69 63 61 6E 20 45 78 70 72 65 73 73 87 01 01
C5	61 0C 4F 07 A0 00 00 00 65 10 10 87 01 01
C6	61 10 4F 07 A0 00 00 01 52 30 10 87 01 01 9F 2A 01 06 61 0D 4F 08 A0 00 00 03 24 10 10 00 87 01 02
C7	61 0D 4F 08 A0 00 00 03 33 01 01 02 87 01 01

2.3.3 eSE or eUICC PPSE Configuration

The Product Provider shall deliver a PPSE test application supporting the update functionality described in 2.3.2. The sequence to update PPSE is performed each time a new kernel is tested.

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2.4 Validation Testing Step

2.4.1 Description

Validation testing is to be performed on a limited number of terminals (up to 6). Access to the list of terminals used for the validation is restricted to Recognised Test Laboratories and EMVCo. Validation testing may be performed manually or by utilizing a robot.

Validation consists of performing a successful transaction with the DUT on all terminals used for validation.

- Transactions are performed with only one of the kernels loaded in the terminal.
- If the test application is not available for the specified kernel, it shall be indicated in the Test Report and another kernel shall be used. If there is no other available kernel, then the relevant tests shall be indicated in the Test Report as non-applicable due to missing test application for the supported kernels.
- The laboratory shall update the PPSE prior to the test to identify the targeted payment application AID. For further information refer to section 2.3.
- If the DUT supports only one execution environment, two samples must be validated.
    - If only one physical sample is used for interoperability, it shall be used for validation. The second sample to validate can be any other from the set of samples provided by the vendor to the laboratory.
    - If two physical samples are used for interoperability (one for type A and other for type B), then both samples shall be used for validation.
- If the DUT supports several execution environments,
    - For the Initial EE, two samples must be validated (with this EE activated).
        - If only one physical sample is used for interoperability, it shall be used for validation. The second sample to validate can be any other from the set of samples provided by the vendor to the laboratory.
        - If two physical samples are used for interoperability (one for type A and other for type B), then both samples shall be used for validation.
    - For each additional EE, one sample has to be validated (with the target EE activated). The sample used for the validation of additional EE shall be one of the samples used for analogue testing for this EE.

    Note: If the activation of the EE can be made by the laboratory, the same physical sample can be used for the validation of 2 different EEs.
- Any digital configuration can be used on the validated samples (digital configuration shall be documented in the Test Report).
- If the DUT supports multiple execution environments, only one execution environment shall be active and/or only one execution environment shall contain active test applications.
- Up to five transactions shall be performed at a single test position (r=0, φ= 0, z=1). No more than five shall be attempted in order to meet the pass conditions. The test position is defined in [CIS] Annex B.
- Only one DUT orientation is used for presentment (θ = zero).
- The RF field shall be activated on the terminal being used before the DUT is presented.

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2.4.2 Pass Criteria

The validation pass criteria is defined as

- At minimum, three out of five transactions per terminal are successful, for all terminals (not necessarily consecutive successful transactions) and for each sample.

The product must pass the validation testing to be eligible to proceed to Interoperability Testing. If the product cannot pass validation testing, regardless of the analogue and digital test results, it shall be considered as a failure and is not eligible to receive a Letter of Approval.

2.5 Interoperability Testing Step

2.5.1 Description

Interoperability testing shall be performed on each terminal from the terminal pool and each supported kernel, using a positioning robot to ensure accuracy of the test positions and DUT movements.

The same sample(s) shall be used by the laboratory to perform the interoperability testing and the full analogue testing: if the same sample is used for tests with Type A and Type B during full analogue testing, the same sample shall be used for interoperability testing. If two different samples are used for tests with Type A and Type B during full analogue testing, both samples shall be used for interoperability testing.

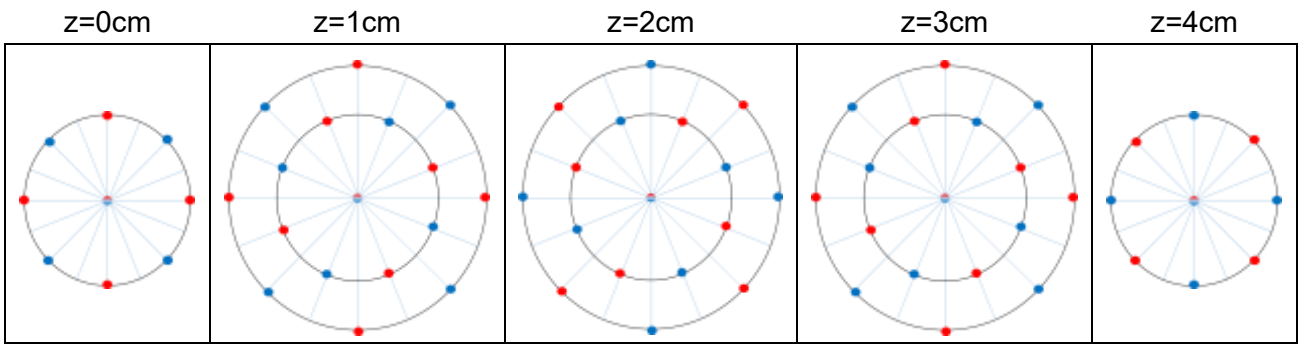
- If the DUT is using UICC, eSE or eUICC, the laboratory shall update the PPSE to identify the targeted AID before each test. For further information refer to section 2.3.
- If the DUT is using HCE (and PPSE is embedded into the HCE application), the laboratory shall enable the application with the targeted AID.

The positioning robot shall perform a loop to present the DUT to the terminal at each position defined in Figure 2-1. Each test position shall be tested once and the test result is either fail or pass. The transaction shall be performed at each position for each payment system kernel required to be tested on the terminal.

The DUT shall be presented to the reader in two orientations: θ = zero and $\theta = -\pi/2$ (Refer to [CIS] Annex B for the definition of θ).

Each orientation θ shall be tested at the test positions and using the protocol indicated in Figure 2-1.

Figure 2-1: Test Positions



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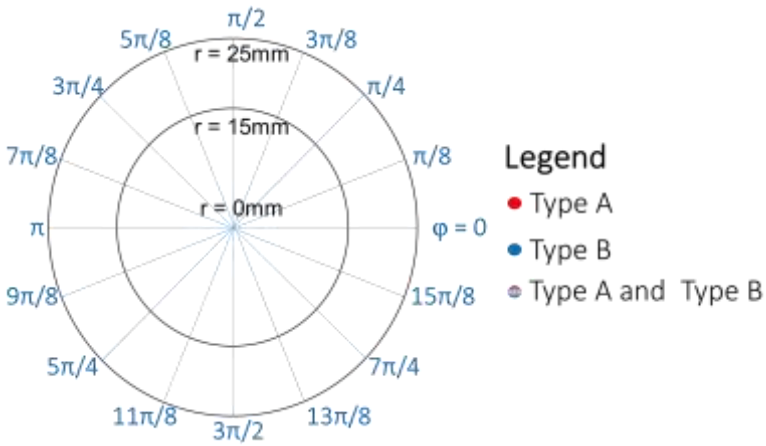

- There are a total of 74 positions defined and 2 orientations. As a result, total 148 transactions shall be performed per terminal for each kernel.
- If the DUT supports only one contactless protocol (Type A or Type B), then all the positions described in Figure 2-1 (Type A and Type B) shall all be tested using the supported contactless protocol.
- If the DUT is supporting Type A and Type B, but the protocol cannot be dictated (configured to work in particular contactless protocol), the positions described above in Figure 2-1 shall be tested with the selection of the protocol determined by the terminal.
- If the DUT configuration supports UICC and ICS indicates that the UICC protocol is overridden by the DUT enforcing a specific protocol configuration independent from the UICC, then a single test UICC can be used.
- In case the test is made using a single protocol type due to one of the above reasons, the five positions with r=0 and θ=0 shall be tested 2 times and all values shall be included in the success rate calculation so that the total sum of weights remains same even though the DUT supports two protocol types or one.
- The DUT shall be held at each test position for 1.5 seconds before being removed.
- The test bench must implement a sufficient delay between each transaction to allow the terminal to restart the transaction for the next test position. The actual delay depends on each terminal. The laboratories are advised to carefully define this value to a minimum to reduce total test duration.
- If the DUT supports multiple execution environments, the interoperability tests have to be performed in accordance with multiple execution environment requirements defined in [MTA ADM].
- If the DUT supports multiple execution environments, only one execution environment shall be active and/or only one execution environment shall contain active test applications.
- If a test application is not available for one of the kernel to be tested, then the test for the terminal/kernel shall be indicated in the Test Report as not applicable due to missing test application for the kernel. A missing test application use case has to be approved by the payment system.

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2.5.2 Pass Criteria

Each test position is designated with a weighted value. The weighted values per terminal are provided in [IT LAB GUID].

The Success Rate of a terminal is defined by the following formula:

- W_i is the weight of particular position i.
- R_i is the Result of the transaction at that particular position i (R_i = 1 if success or R_i=0 if transaction failed).
- TWV (Total Weighted Value) is the sum of all weighted values at all test positions TWV = ΣW_i.
- TR is the test result for one terminal using one kernel TR = $\Sigma (R_i * W_i)$.
- Success Rate = TR/TWV.

Note 1: The success rate is calculated using the sum of type A and type B weighted values. It is calculated for each terminal/kernel/orientation.

The pass criteria for interoperability testing is defined below:

1. The Success Rate shall be greater than or equal to 83% for each terminal/kernel/orientation regardless of protocol type.
2. The transaction shall pass successfully on at least 50% of the positions with z=0cm for each protocol type, for a given terminal/kernel/orientation.
3. Criteria 1 and 2 must be met on every terminal in the pool.

Note 2: The pass criteria and weighted values are subject to change over time.

Note 3: For the second pass criteria, if the DUT supports only one protocol type, the laboratory has to calculate two % results on 5 positions for the z=0 criteria (to have equivalent results as a DUT supporting two protocol types).

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3 Requirements For Testing Laboratories

3.1 Testing Conditions

The DUT shall be tested in the following conditions:

- Without the use of any external RF device that could have an impact (e.g. RF spy tool).
- The DUT must not be charging during testing. If necessary, the DUT can be recharged between test sessions, or the battery can be replaced with a battery of the exact same model (if the NFC - RF antenna is not on the battery). When multiple batteries are used during the test, the laboratory shall clearly indicate which battery was used for which test session and which terminal/kernel.
- The DUT is switched on.
- If the validation or interoperability fails for a terminal/kernel, the laboratory shall verify that the DUT battery level is not in power saving mode. If this is the case, the test shall be run again with a recharged battery.
- If the validation or interoperability fails for a terminal/kernel, and other terminals (not under test) are active in the vicinity, the laboratory shall turn them off to verify that they don't have any influence on the terminal/kernel under test.
- If supported by the DUT, Bluetooth, GPS and WiFi shall be disabled or turned off.
- All contactless functionalities of the DUT shall be enabled (e.g. Type A, Type B, NFC-F, P2P, R/W Mode, etc.). Note: HCE shall not be activated if either SWP UICC, eSE or eUICC is the execution environment under test.
- If the DUT requires the display on to enable the contactless technologies and if it is not possible to deactivate the display off timer, the laboratory may install an application that will maintain the display on.
- The PPSE shall only refer to one payment application.
- The DUT must be provided with a Reference Mark representing the reference test point.
- The terminal shall be connected to external power supply during the test session.
- The terminal may be connected to a PC.
- The interval between powering of the terminal and the start of the actual testing is a minimum of 5 minutes (warmup time).
- The default configuration of a slide, flip or swivel DUT shall be open as the way of normal mobile usage.

3.2 Forms

The Product Provider must submit a completed ICS (conforming to [L1ICS]) to the test laboratory prior to Validation and Interoperability Testing and as part of Mobile Product Level 1 Type Approval.

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3.3 Normal Temperature and Humidity

The normal temperature and humidity conditions for Validation and Interoperability Testing shall be any convenient combination of temperature and humidity within the following ranges:

- Temperature between: 23°C ± 3°C
- Relative humidity: 40 % to 60 %.

When it is impractical to carry out tests under these conditions, a note to this effect, stating the ambient temperature and relative humidity during the tests, shall be added to the Test Report.

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4 Positioning Conventions

4.1 Terminal Positioning Conventions

4.1.1 Terminal Z Axis

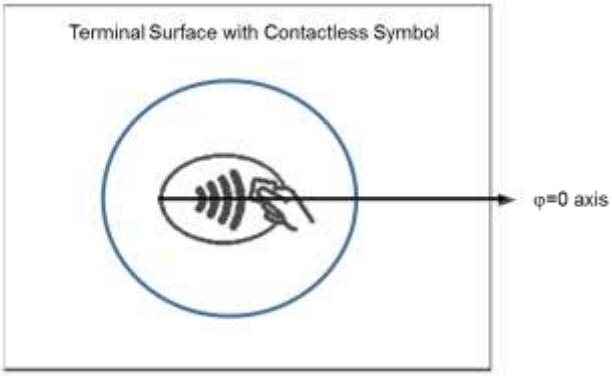
The following characteristics are introduced to support the positioning convention:

- The landing plane is the area identified on the terminal, where a laboratory should position the DUT to obtain a successful reading.
- The centre of the landing plane is indicated by the contactless symbol.
- The z axis of the terminal is the axis going through the centre of the landing plane, orthogonal to the landing plane or to the tangent of the landing plane and directed outwards from the terminal surface.

4.1.2 φ = 0 Axis

The φ = 0 axis of the terminal is indicated by the Contactless Symbol.

Figure 4-1: φ = 0 axis on the Test Terminal



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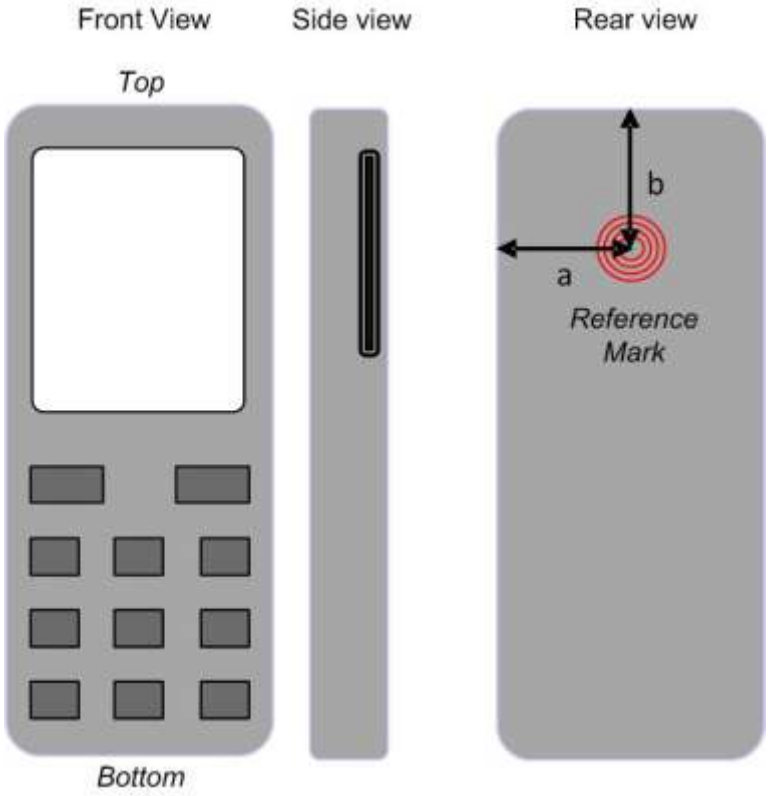
4.2 DUT Positioning Conventions

DUT positioning conventions used in this document are compliant with [CIS] Annex B.

An example of Device Under Test is shown in below diagram (refer to [L1ICS] part L1A.6.8 or L1A.6.10 for the Reference Mark position):

Figure 4-2: DUT Reference Mark



4.2.1 DUT Z Axis

The following characteristics are introduced to support the positioning convention:

- The presentation plane is the outer part of the DUT that is presented over the landing plane of the terminal. The outline of the presentation plane shall be indicated on the DUT.
- The “Reference Rark” is the point to be presented over the centre of the landing plane of the terminal and it shall be indicated on the DUT.
- The z axis of the DUT is the axis going through the Reference Mark, orthogonal to the presentation plane or to the tangent of the presentation plane and directed outwards from the DUT surface.

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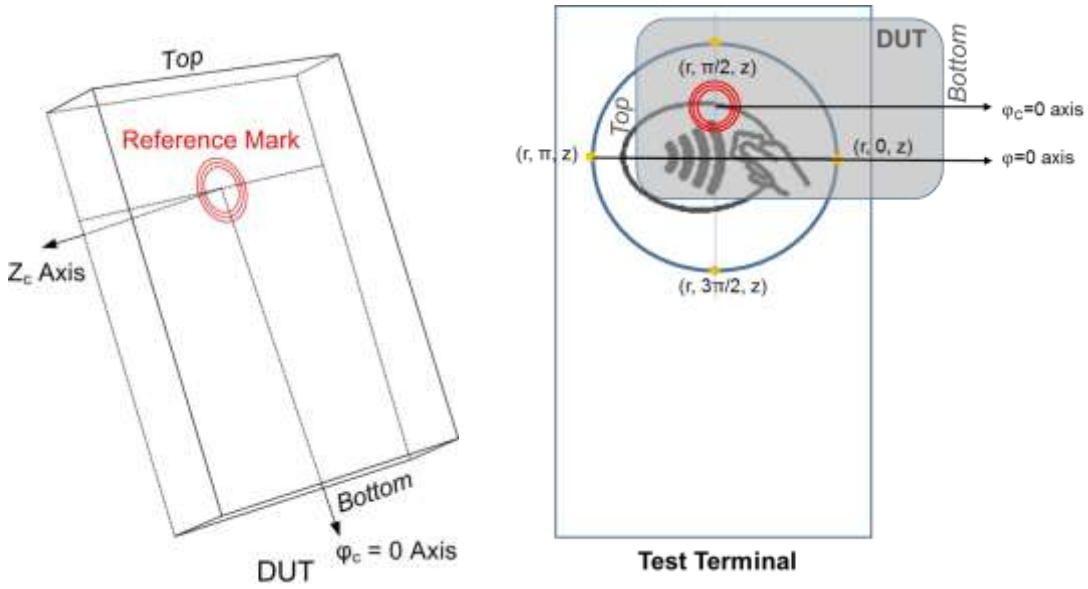
4.2.2 DUT orientation θ=0

The DUT orientation θ=0 is defined by:

- The DUT φc = 0 axis must be aligned with the φ = 0 axis of the test terminal.
- The top of the DUT must be placed on the test terminal (r, π, z) position side as shown in below diagram.

Figure 4-3: DUT θ = 0 Definition



The following pictures present examples of position and orientation of the DUT when φ and θ vary.

Figure 4-4: Positioning for θ = 0, φ = π/2, r ≠ 0

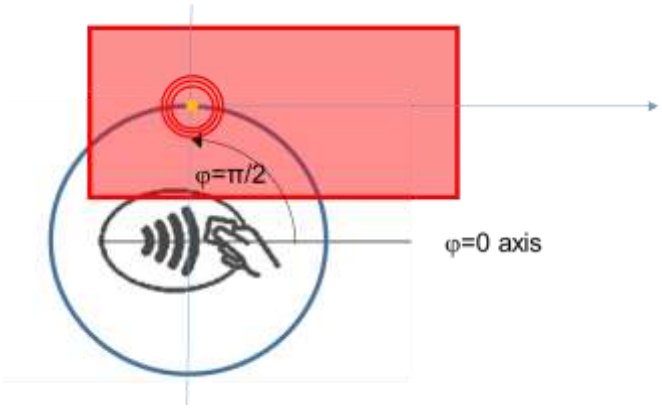


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Figure 4-5: Positioning for $\theta = 0$, $\varphi = 3\pi/2$, $r \neq 0$

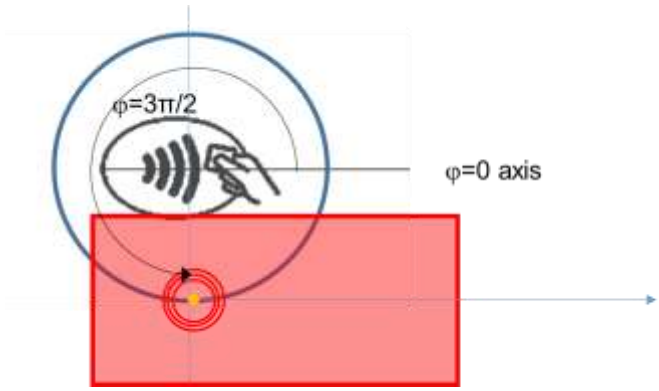
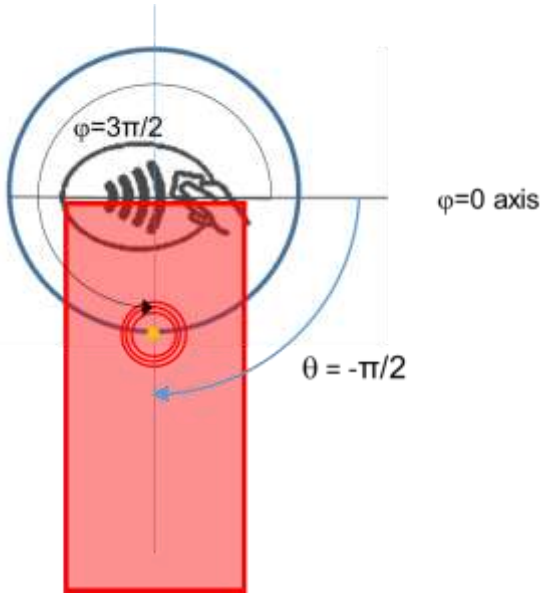


Figure 4-6: Positioning for $\theta = -\pi/2$, $\varphi = 3\pi/2$, $r \neq 0$

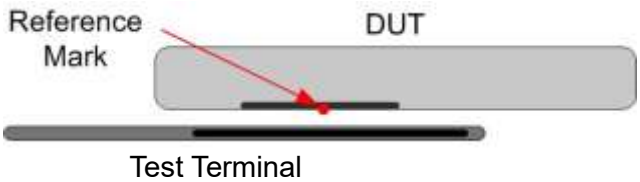


4.3 DUT Presentation Rules

In order to avoid DUT positioning inconsistencies between laboratories, the following rules must be followed:

- The presentation plane of a DUT shall always be parallel to the landing plane of the test terminal.
- The DUT must be presented to the Test terminal on the side of the Reference Mark.

Figure 4-7: DUT position on the Test Terminal



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5 Requirements for the Positioning System

5.1 Introduction

The Positioning System used for the Interoperability Testing step, shall use a robot and shall be programmed to carry out specific actions to place the DUT in the various test positions, repeatedly, without variation and with a high degree of accuracy. These actions are determined by programmed routines that specify the direction, acceleration, velocity, deceleration, and distance of a series of coordinated motions.

The robot system shall consist of a fixed chassis and at least one arm, for holding and moving small objects. Three axis are required to reach any point in space.

The robot shall be connected to a communication port of a computer in order to be correctly configured with dedicated software.

5.2 General Requirements

- The positioning system shall be able to position the DUT attached to its moving arm at all test positions defined in 2.5.1.
- The terminal is securely positioned.
- When the DUT is presented to the terminal, there shall not be any metallic object in the close vicinity of the DUT, to ensure that the electro-magnetic field is not disturbed. A distance of at least 15cm shall be kept between the DUT and any metallic object pertaining to the positioning system.
- A distance of at least 15cm shall be kept between the terminal and any metallic object pertaining to the positioning system.
- A distance of at least 30 cm shall be kept between the terminal and any other switched on terminal.
- When the DUT is presented to the terminal, the presentation plane of a DUT shall always be parallel to the landing plane of the test terminal.

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5.3 Positioning Requirements

5.3.1 General Requirements

- The positioning system shall be programmable so that the position of any point relative to the origin shall be programmable with at least three coordinates (r, φ, z).
- Whenever the surface of the terminal is not flat, the Landing Plane shall be defined as the plane that is tangent to the centre of the Contactless Symbol.
- The tolerance on the parallelism between the DUT presentation plane and the landing plane shall not exceed 5°.

5.3.2 Untestable Positions at z=0cm

It may not be possible to place the presentation plane of the DUT against the landing plane (z=0cm) of the terminal due to the shape of the terminal and/or the DUT. In this case, the testing for positions at z=0cm is performed at the closest achievable level to 0cm. This new position will be identified as z0. z0 values are defined for each terminal in [IT LAB GUID].

If the terminal offset (z0) is greater than 7mm, then all the positions at z=0cm shall not be tested. The following rules apply in this case:

- The weights associated to z=0cm are not included into the calculation of TR and TWV (see section 2.5.2).
- The second pass criteria (50% pass at z=0cm – see section 2.5.2) is evaluated at z=1cm, inner ring (r=0 mm or r=15mm) instead of z=0cm.

Note 1: Unless specified differently in [IT LAB GUID], z0 position is unique for a given terminal whatever i) the shape of the (mobile phone) DUT, ii) the orientation of the DUT, iii) the position of the DUT.

If [IT LAB GUID] indicates that z0 is different for each orientation of the DUT, then the rules described in sections 5.3 are applied specifically for each orientation and z0.

Note 2: If test is possible at z=0cm, for all positions of a given DUT orientation of a specific DUT, then the z0 rule does not apply for this orientation (and test is made at z=0cm). If test is not possible for at least one position at z=0cm for a given DUT orientation, then the z0 rule does apply for this orientation (and test is made at z=z0).

It shall be indicated in the Test Report if z0 was used (for each terminal and DUT orientation).

Note 3: If the DUT is a non-card L1 form factor, please contact EMVCo to know if Note 2 applies or not.

5.3.3 DUT Offset

The DUT may add an additional offset because of its shape (the camera for instance may create additional offset).

- If the DUT's offset plus the terminal offset (z0) is less or equal to 7mm, then z0 plus DUT offset shall be used for interoperability testing at z=0cm.
- If the DUT's offset plus the terminal offset (z0) is greater than 7mm, then all the positions at z=0cm shall not be tested. The following rules apply in this case:
    - The weights associated to z=0cm are not included into the calculation of TR and TWV (see section 2.5.2).

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- The second pass criteria (50% pass at z=0cm – see section 2.5.2) is evaluated at z=1cm, inner ring (r=0 mm or r=15mm) instead of z=0cm.

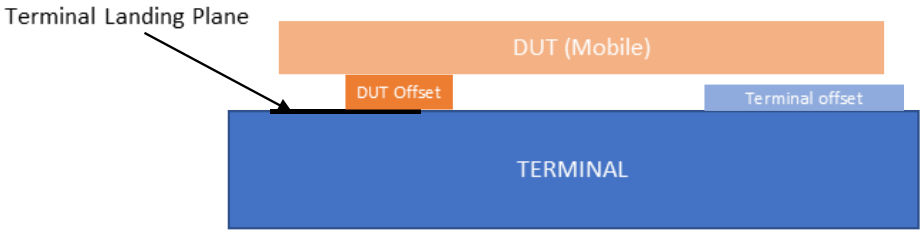
- DUT offset plus terminal offset (z0) shall be documented in the Test Report.

Note 1: the DUT offset shall only be considered (for all positions and all DUT orientations) when it prevents testing at 0cm or z0 for at least one position on a given DUT orientation. For instance, if a camera creates an offset on the presentation plane of the DUT but it does not prevent testing at 0cm or z0 for all positions and DUT orientations, the DUT offset shall not be taken into account.

Note 2: if the DUT offset is higher than the terminal offset and if it prevents testing at 0 or z0 but is never facing the terminal part that creates the terminal offset for any of the test positions and DUT orientations, then the Terminal offset shall not be taken into account and only the DUT offset shall be used for interoperability testing.

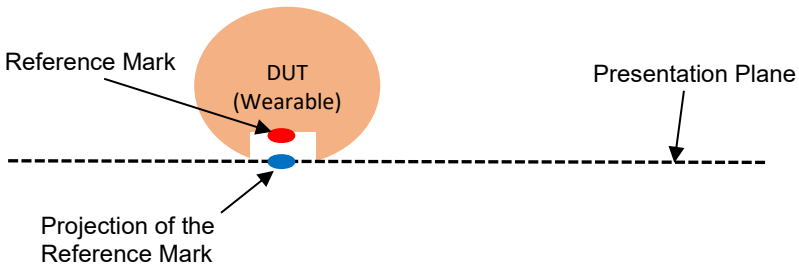
If the DUT offset is facing the terminal part that creates the terminal offset for at least one of the test positions on a given DUT orientation, then the DUT offset shall be taken into account and added to the terminal offset for all positions and all DUT orientations.

Figure 5-1: DUT offset is higher than Terminal offset



Note 3: The Presentation Plane of Wearable Products shall be defined and documented in the ICS so that all positions at z=0cm (or z0 for terminals that do not support testing at 0cm) are testable during Interoperability testing: ie, DUT offset does not apply to Wearable Product.

It may happen that the Reference Mark is not on the presentation plane due to the shape of the Wearable Product. In that case, a projection of the Reference Mark on the presentation plane is used to define the origin. This projection shall be described in the ICS. See [L1ICS] part L1A.6.8 or L1A.6.10).



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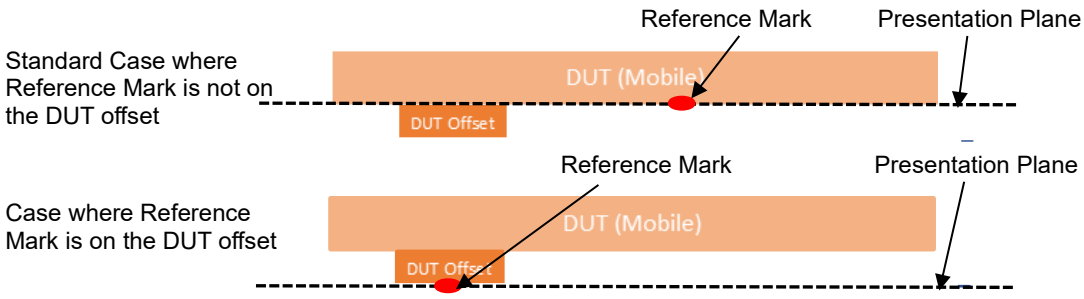
Note 4: if the Reference Mark of a non Wearable DUT is located on the part of the DUT that creates a DUT offset (on the camera for instance) then the presentation plane of the DUT shall be defined to include the Reference Mark and no offset due to the DUT offset shall be applied during testing.



Note 5: if the DUT is a non-card L1 form factor, please contact EMVCo to know if the Notes 1 to 4 apply or not.

5.3.4 Untestable Positions at z=1cm

It may not be possible to place the presentation plane of the DUT at z=1cm of the landing plane of the terminal due to the shape of the terminal and/or the DUT. In this case, the testing for positions at z=1cm is performed at the closest achievable level to 1cm.

- If the DUT's offset plus the terminal offset (z0) is greater than 10mm but is less or equal to 17mm, then z0 plus DUT offset shall be used for interoperability testing. This also applies to the testing of the second pass criteria (50% pass at z=0cm – see section 2.5.2) that is evaluated at z0 plus DUT offset, inner ring (r=0 mm or r=15mm).
- If the DUT's offset plus the terminal offset (z0) is greater than 17mm, then all the positions at z=1cm shall not be tested. The following rules apply in this case:
    - The weights associated to z=1cm are not included into the calculation of TR and TWV (see section 2.5.2).
    - The second pass criteria (50% pass at z=0cm – see section 2.5.2) is evaluated at z=2cm, inner ring (r=0 mm or r=15mm) instead of z=0cm.
- DUT offset plus terminal offset (z0) shall be documented in the Test Report

5.4 Movements Requirements

- The DUT shall be presented in a vertical movement (up/down) with no horizontal movement in the terminal operating volume (the vertical and horizontal movements are defined using the landing plane of the terminal as a reference).
- The positioning system shall move the DUT during the vertical movement, excluding the start and end, at a speed of 1 m/s.
- The DUT shall not be moving in the operating volume for more than 100ms. Consequently, the positioning system shall decelerate the DUT at the end of the vertical movement at a rate of minimum 8 m/s².

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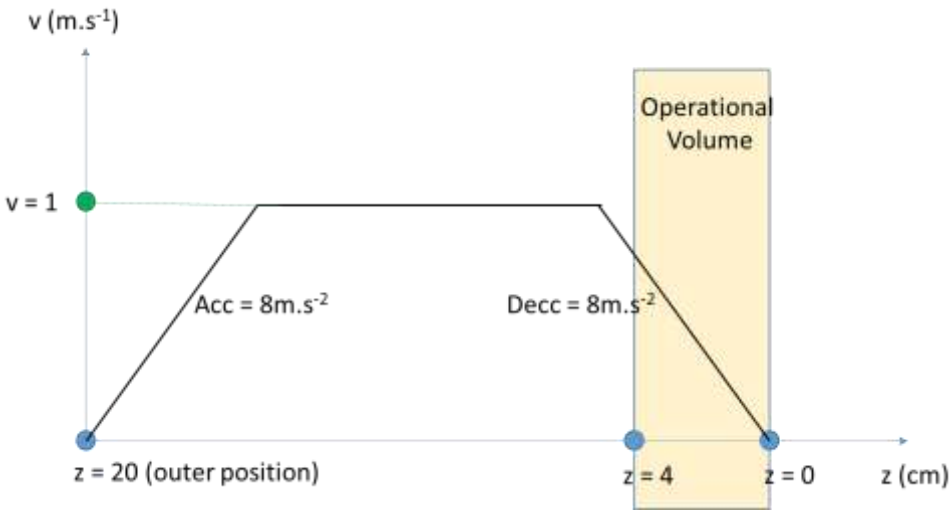
The expected movement is represented in the Figure 5-2 below.

Figure 5-2: DUT movement to position for target z=0cm



5.5 Physical Alignment of DUT, Terminal and Robot

5.5.1 Alignment of the Z Axis

The purpose of this procedure is to align the z axis of the robot and the z axis of the terminal. An automated process using laser measurement and automatic placing is possible.

Step	Actions
1	• Identify the z axis of the positioning system. • Prepare the positioning system to ensure the z axis is visible.
2	• Identify the z axis of the terminal and if possible, make the z axis of the terminal visible. • Identify the φ=0 axis of the terminal.
3	• Mount the terminal and the positioning system next to each other so that the z axis of the positioning system is nearly parallel to the z axis of the terminal. • Verify that the positioning system will be able to cover the operating volume of the terminal.
4	• Move the terminal and the positioning system so that the z axis of the positioning system is parallel to the z axis of the terminal.
5	• Mechanically secure the terminal and the positioning system in this position.
6	• Secure the z axis of the positioning system in this position.

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5.5.2 Mounting DUT on the Positioning system

The purpose of this procedure is to ensure that the orientation of the DUT conforms to the requirements listed in 2.5.1, with the value of the θ being zero or -π/2

Step	Actions
1	Fix the DUT to the arm of the positioning system to fulfil the following conditions: The z axis of the DUT is parallel to the z axis of the positioning system and points in the opposite direction. The angle between the φ=0 axis of the terminal and the φ_c=0 axis of the DUT is 0 (or –π/2).

5.5.3 Installation of the Origin (0,0,0)

The purpose of this procedure is to ensure that at position (0,0,0), the Reference Mark of the DUT, is as close as possible to the centre of the contactless symbol of the terminal.

Step	Actions
1	Program the positioning system to move the DUT so that the Z axis of the terminal crosses the centre of the DUT. The centre of the DUT is defined in the [L1ICS] L1A 5.9. It is usually at the centre of the contactless logo on the phone cover.
2	Reset the coordinating system for the positioning system so that r = 0 for this position.
3	Carefully let the positioning system move the DUT down until it is just one vertical increment away from touching the terminal landing plane. (The DUT shall not touch the terminal landing plane). This can be verified by placing a sheet of paper between the DUT and the terminal such that it is still possible to move the paper).
4	Raise the DUT again by 0.5 mm (to compensate for the tolerance in positioning of the positioning system).
5	Reset the coordinating system of the positioning system so that the z axis for this position is 0 (except if DUT cannot be positioned at z=0: see 5.3).

5.6 Vertical Presentation of the DUT

This section describes how the positioning system shall be programmed to:

1. Bring the DUT into the Operating Volume of the terminal.
2. Keep the DUT at the programmed position for the required time defined in section 2.5.1.
3. Remove it again from the Operating Volume of the terminal.

For each of the programmed positions, the positioning system will begin the presentation from a position with the same values of (r, φ) as the programmed position, but with $z_{initial}$ = z + 200mm (the outer position).

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The positioning system shall move the DUT along a straight line parallel to the z axis until the programmed position is reached. The positioning system shall then maintain the DUT in that programmed position for 1.5 second, and subsequently withdraw it again along a line parallel to the z axis.

When moving the DUT to or from the programmed position, the positioning system shall be programmed to move the DUT in a precise way compliant with the movement requirements described in section 5.4. This applies to acceleration, move and deceleration of the DUT along the z axis. This is to ensure timing repeatability of the presentation.

5.7 Tolerance

This section describes the tolerance for the various measurements:

Item	Tolerance value
Duration of stay at a test position before the DUT is positioned back to the outer position	+/- 0.1s
Position (r, φ, z). Tolerance in any direction between the programmed position and the actual position obtained	+/- 0.5mm
Vertical Acceleration or Deceleration	+/- 1.5m/s²
Vertical Speed	+/- 0.1 m/s
Orientation of the DUT (θ)	+/- 3°
Deviation between z axis of the positioning system and Z axis of the terminal	+/- 3°
Deviation between the landing plane of the terminal and the presentation plane of the DUT	+/- 5°
Deviation between z axis of the positioning system and z axis of the DUT	+/- 3°
Deviation between φ=0 axis of the positioning system and φ_c=0 axis of the DUT	+/- 3°
Moving time in the operating volume (see 5.4)	+/- 10ms

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Appendix A: PPSE Personalisation Script

The EMVCo PPSE test applet does not require security for mutual authentication.

The APDU commands below defines the configuration for each Payment System.

C2 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU:
80 E2 80 00 29 50 01 26 6F 24 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 12 BF 0C 0F 61 0D 4F 08 A0 00 00 00 04 10 10 11 87 01 01
Response APDU:
90 00

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C3 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU:
80 E2 80 00 28 50 01 25 6F 23 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 11 BF 0C 0E 61 0C 4F 07 A0 00 00 00 03 10 10 87 01 01
Response APDU:
90 00

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C4 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU:
80 E2 80 00 3B 50 01 38 6F 36 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 24 BF 0C 21 61 1F 4F 08 A0 00 00 00 25 01 09 01 50 10 41 6D 65 72 69 63 61 6E 20 45 78 70 72 65 73 73 87 01 01
Response APDU:
90 00

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C5 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU:
80 E2 80 00 28 50 01 25 6F 23 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 11 BF 0C 0E 61 0C 4F 07 A0 00 00 00 65 10 10 87 01 01
Response APDU:
90 00

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C6 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU: 80 E2 80 00 3B 50 01 38 6F 36 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 24 BF 0C 21 61 10 4F 07 A0 00 00 01 52 30 10 87 01 01 9F 2A 01 06 61 0D 4F 08 A0 00 00 03 24 10 10 00 87 01 02
Response APDU:
90 00

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C7 PPSE Personalisation Script

Select Applet, PPSE
Command APDU:
00 A4 04 00 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 00
Response APDU:
6F xx … xx 90 00

Initialize Update
Command APDU:
80 50 00 00 00
Response APDU:
90 00

External Authenticate
Command APDU:
00 82 00 00 00
Response APDU:
90 00

Store Data
Command APDU:
80 E2 80 00 29 50 01 26 6F 24 84 0E 32 50 41 59 2E 53 59 53 2E 44 44 46 30 31 A5 12 BF 0C 0F 61 0D 4F 08 A0 00 00 03 33 01 01 02 87 01 01
Response APDU:
90 00

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